Ethics, Privacy, and Security in Data Science and AI (CST4235)

Week 2

(15 Credits)

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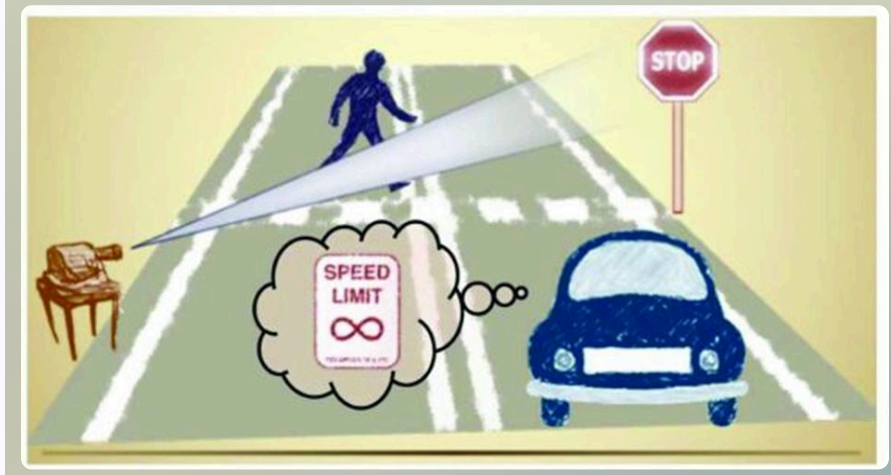
Trustworthy AI - 7 Requirements

- **Human agency & oversight** → AI should support, not replace, human decision-making.
  Example: *Doctors make final calls, not just the AI system.*
- **Technical robustness & safety** → AI should not crash or cause harm.
  Example: *Self-driving cars must work safely in all conditions.*
- **Privacy & data governance** → Protect personal data.
  Example: *Your health app should not sell your medical history.*
- **Transparency** → Clear explanations of how AI makes decisions.
  Example: *Why was my loan rejected?*
- **Diversity, non-discrimination & fairness** → Avoid bias.
  Example: *Hiring AI should not favour one gender or ethnicity.*
- **Societal & environmental well-being** → Benefits society as a whole.
  Example: *AI for clean energy, not just profit.*
- **Accountability** → Someone must be responsible for AI outcomes.
  Example: *Who is liable if an AI makes a mistake?*

AI Robustness and Safety

Today's Agenda

- Use of AI in Safety-Critical Applications
- Definition of Robustness
- Adversarial Examples
- Enabling Safe and Robust AI
- Specification-Driven Machine Learning
- Adversarial Defence Strategies
- Distribution Shifts: Real-World Examples
- Making Models More Robust to Distribution Shifts



AI in Safety-Critical Domains

Where AI is used

- Self-driving cars → split-second decisions
- Robot-assisted surgery → life-saving operations
- Drug discovery → predictions for critical medicines
- Diagnostics → detect diseases without human input

Reminder

- Innovated 5 - 10 years ago
- Still not in daily life

Examples

- Tesla & Waymo cars → tech works, but trust & regulation lag
- AI cancer detection → works in labs, not yet hospitals

Bottom Line

Not just about building AI – it must be **robust, safe & trusted**.



Robustness

- Ensures AI behaves **acceptably** under:
  - Abnormal / unforeseen events
  - Unusual / high-impact situations
  - Adversarial attacks - e.g., stickers on a STOP sign fool AI into reading it as a **Speed Limit 45** sign
- **Why?**
  - Weak models = financial loss
  - Non-robust models = safety risks

Example:
Self-driving car → must recognise stop sign even if partly covered or vandalised.

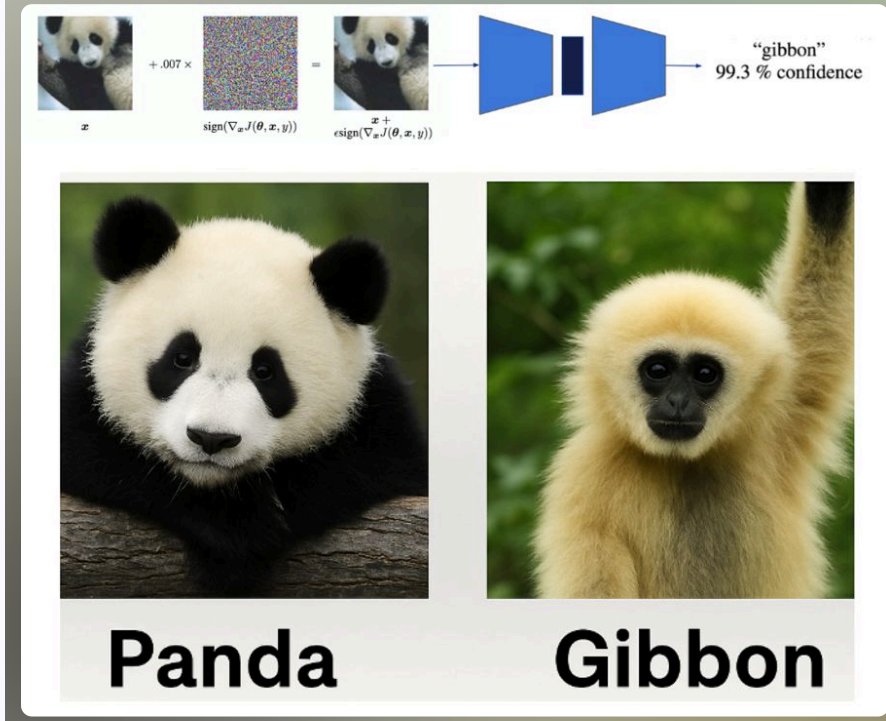


Challenges to Achieving Robustness in ML Models

- **Adversarial attacks** → A few stickers on a STOP sign make an AI think it is a speed-limit sign.
- **Distribution shifts** → A medical AI trained on UK patients fails on data from Asia.
- **Noisy data** → A voice assistant struggles with background noise or unclear speech.
- **Generalization** → A model that works in the lab fails when used in the real world.
- **Model complexity**
  - Too simple → misses patterns (**underfitting**)
  - Too complex → memorises data (**overfitting**)
  - **Goal → balance**

Adversarial Perturbation (Concept & Example)

- In 2013, researchers showed that neural networks can be tricked by small, invisible changes.
- Original input: Panda → correctly classified as *“panda”* with 57.7% confidence.
- Add tiny, carefully chosen noise → image still looks like a panda to humans.
- Neural network misclassifies: *“gibbon”* with 99.3% confidence.



Implications of Adversarial Attacks

- Misclassification seems harmless (e.g., panda → gibbon).
- An attacker can search for a tiny change (perturbation) that shifts the model's prediction. With the right perturbation the model can be made to output any chosen label.

Example: By tweaking a few pixels on a stop sign until the classifier's top prediction flips from *Stop* (72%) to *Speed 45* (96%) – the image still looks unchanged to humans.



Blackbox Adversarial Example

- **Left (a: Original image):** The washing machine is classified correctly as “washer” with 53% probability.
- **Right (b: Adversarial example):** After altering the resolution (a subtle perturbation), the classifier is fooled. It no longer sees it as a washer – instead, it outputs “safe” (37%) and “loudspeaker” (24%).

Even very small or indirect changes (like lowering resolution, which a human would still recognise as a washing machine) can trick a model into completely misclassifying an image.

Figure 3. An example of a black-box adversarial example. The original image (a) is classified as “washer” with 53 percent probability. The image is altered by reducing its resolution to create an adversarial example (b), which is classified as “safe” with 37 percent and as “loudspeaker” with 24 percent probability.



a) Original image b) Adversarial example

Source: Alexey Kurakin, Ian Goodfellow, and Samy Bengio, “Adversarial Examples in the Physical World,” arXiv [cs.CV] (February 11, 2017), preprint, https://arxiv.org/abs/1607.02533.

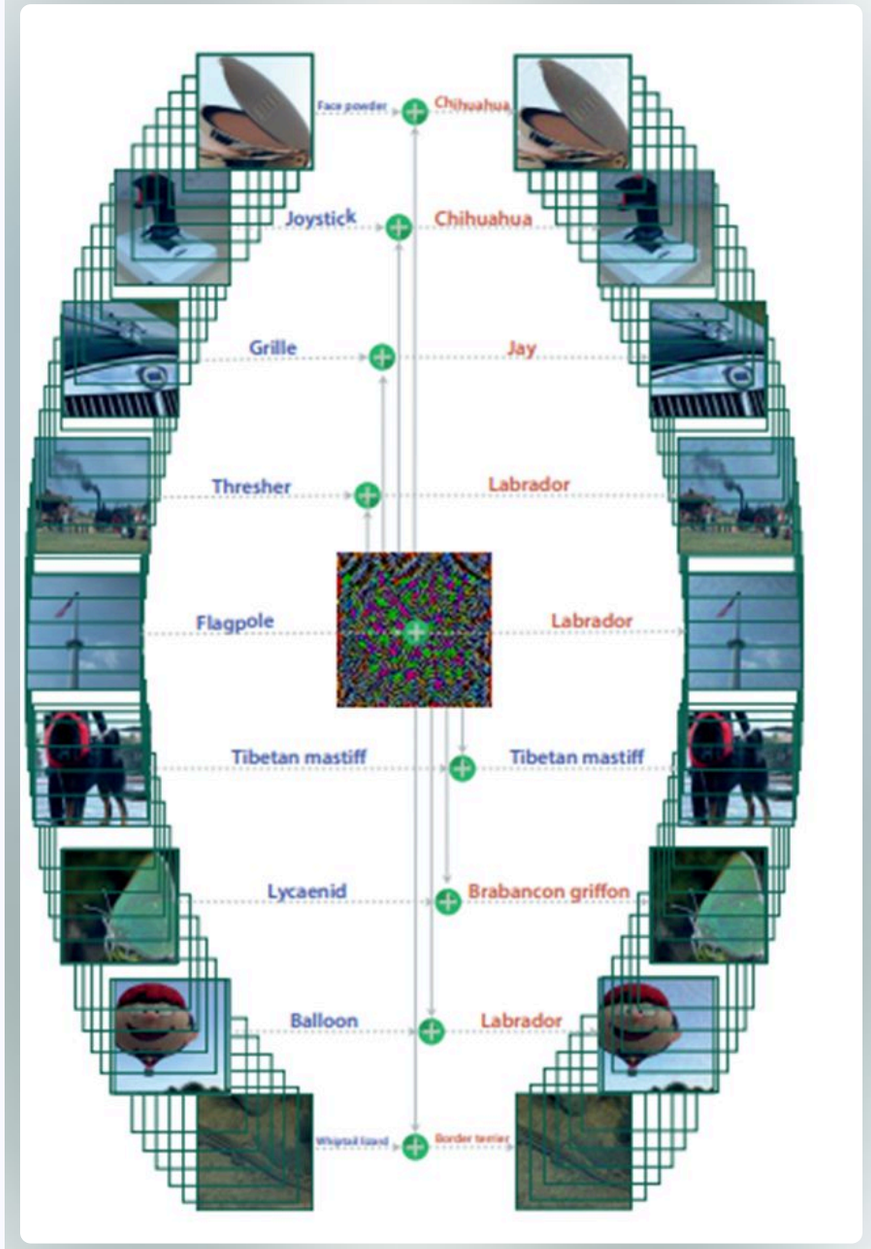
Universal Adversarial Perturbation (UAP)

- An adversarial perturbation is normally made **for one specific image** (e.g., a panda image is altered to look like a gibbon to the model).
- In 2017, researchers showed something more powerful: you can create **a single noise pattern** that works on **many different images**, not just one.
- It is like a "magic sticker" or "universal trick" – once you place it on any image, the AI model gets confused and misclassifies.

Example: *The same noise pattern, when added to pictures of:*

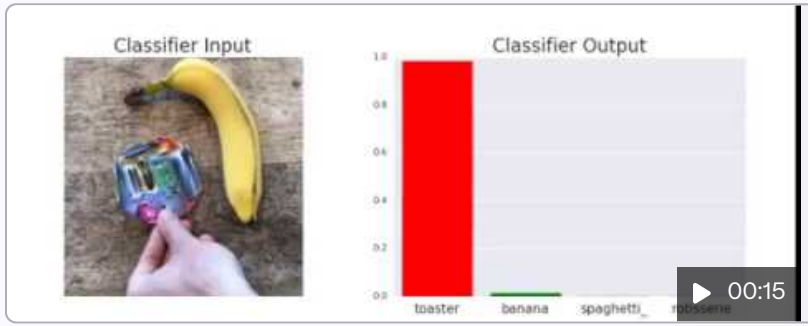
  - *a dog → misclassified as a cat*
  - *a car → misclassified as an airplane*
  - *a face → misclassified as something else*

Even though the human eye barely notices the noise, the model's output **flips for many different inputs**.



Moosavi-Dezfooli, S. M., Fawzi, A., Fawzi, O., & Frossard, P. (2017). Universal adversarial perturbations. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1765-1773).

Adversarial Patch Example



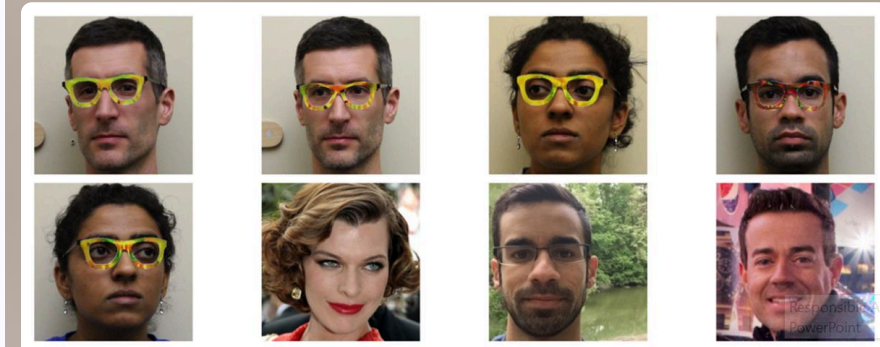
YouTube

Adversarial Patch

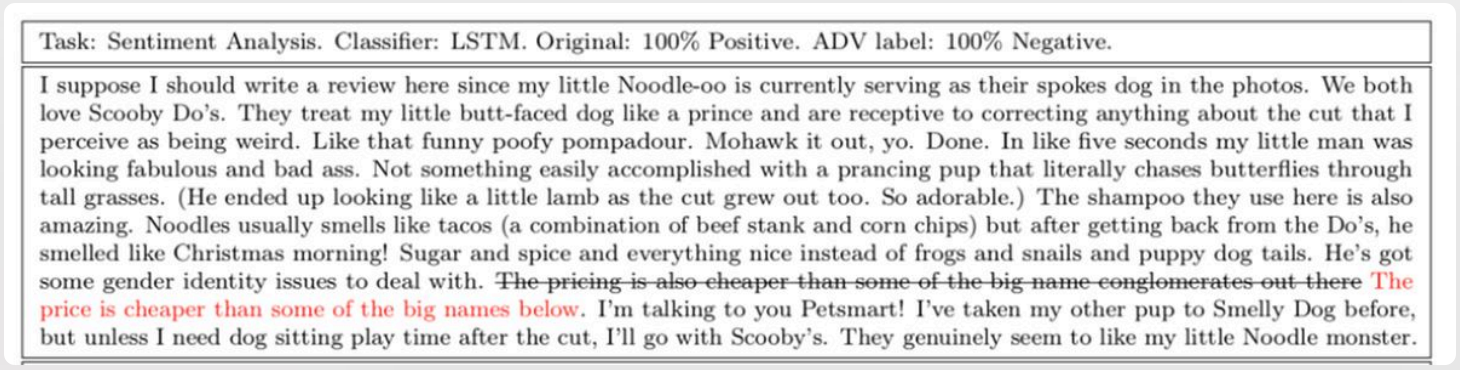
A real-world attack on VGG16, using a physical patch generated by the white-box ensemble method described in the Adversarial Patch paper: http://arxiv.org/abs/1712.09665 Our adversari...

Adversarial Example - Image

- In 2016, researchers at Carnegie Mellon showed that **something as simple as wearing colourful glasses** could trick facial recognition systems.
- The AI mistakenly identified ordinary people as **celebrities** just because of these glasses.
- **Why?**
  - AI models rely heavily on **patterns of pixels**.
  - The colourful glasses created **unnatural patterns** that the AI thought were **unique features of a celebrity's face**.
- For humans, the glasses are just glasses. For the AI, they became a **signal strong enough to override the real face** → leading to **misclassification**.



Researchers at Carnegie Mellon University discovered that by donning special glasses, they could fool facial recognition algorithms to mistake them for celebrities (Source: http://www.cs.cmu.edu)



Task: Sentiment Analysis. Classifier: LSTM. Original: 100% Positive. ADV label: 100% Negative.
I suppose I should write a review here since my little Noodle-oo is currently serving as their spokes dog in the photos. We both love Scooby Do's. They treat my little butt-faced dog like a prince and are receptive to correcting anything about the cut that I perceive as being weird. Like that funny poofy pompadour. Mohawk it out, yo. Done. In like five seconds my little man was looking fabulous and bad ass. Not something easily accomplished with a prancing pup that literally chases butterflies through tall grasses. (He ended up looking like a little lamb as the cut grew out too. So adorable.) The shampoo they use here is also amazing. Noodles usually smells like tacos (a combination of beef stank and corn chips) but after getting back from the Do's, he smelled like Christmas morning! Sugar and spice and everything nice instead of frogs and snails and puppy dog tails. He's got some gender identity issues to deal with. ~~The pricing is also cheaper than some of the big name conglomerates out there~~ The price is cheaper than some of the big names below. I'm talking to you Petsmart! I've taken my other pup to Smelly Dog before, but unless I need dog sitting play time after the cut, I'll go with Scooby's. They genuinely seem to like my little Noodle monster.

Text-based Adversarial Examples (Paraphrasing Attacks)

- In **2019**, researchers from **IBM Research, Amazon, and the University of Texas** showed that text classifiers (e.g., spam filters, sentiment detectors) can be tricked by carefully modified text.

Sentiment Analysis

- The model is trained to classify text as **Positive** or **Negative** sentiment.
- **Original review** → The whole review is full of **praise**:
  - Words like *"love," "adorable," "fabulous," "amazing," "sugar and spice and everything nice."*
  - Because of this, the classifier gave it **100% Positive**.

Adversarial Change (Paraphrasing Attack)

- A **tiny change** was introduced to one sentence:
  - **Original:** *"The pricing is also cheaper than some of the big name conglomerates out there."*
  - **Changed to:** *"The price is cheaper than some of the big names below."*

Why does this fool the model?

- To a **human reader**, the meaning is basically the same.
- But the **model's attention** shifts:
  - It sees the phrase ***"cheap" + "big names"*** in a new context.
  - The model mistakenly interprets this as a **negative sentiment** (maybe associating "cheap" with "low quality" in its training data).

Enabling Safe and Robust AI

1. Vanilla Training (default way we train models)

Train only on the data we have.

Problem: if data is biased or limited, the model will copy those flaws.

Example: If the training set mostly shows men as doctors and women as nurses → the model will repeat this stereotype.

2. What Goes Wrong

Biased models → unfair predictions.

Non-robust models → fail when data changes slightly.

Example: A spam filter trained mostly on English may fail badly when it sees mixed English-Spanish emails.

Specification-Driven Training (the solution)

We add rules/specifications to guide the model during training.

Think of them as safety checks.

Examples of rules:

A dosage prediction system must never suggest a negative dosage.

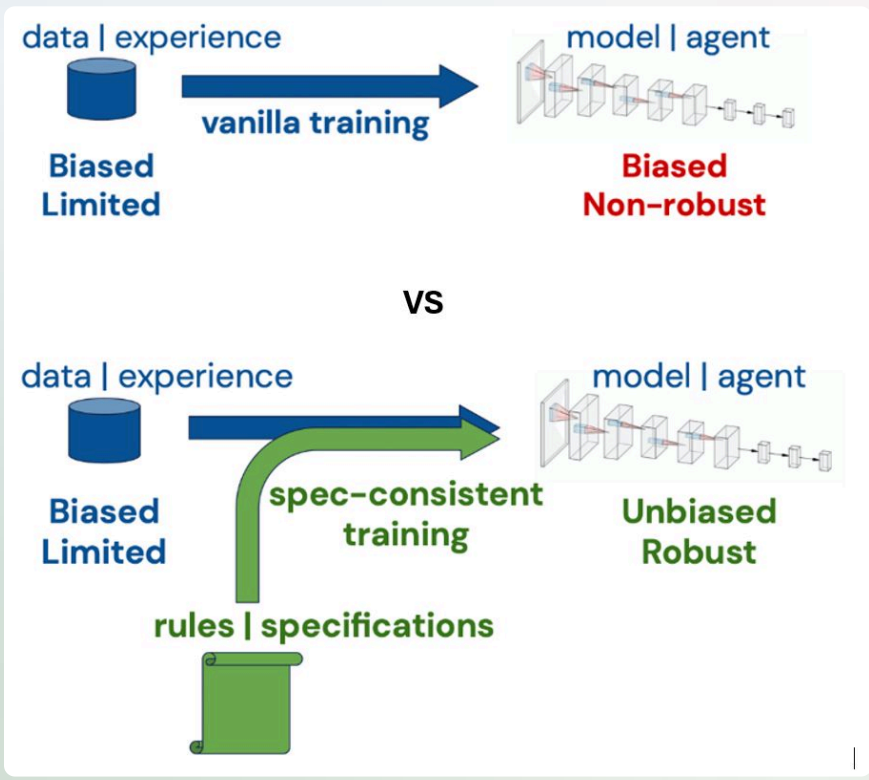
Probabilities must always add up to 1.

Credit approval system must treat male and female applicants equally.

Outcome

Without rules (vanilla): Biased + non-robust.

With rules (spec-driven): Fair + robust + reliable.



Robustness to Irrelevant Changes

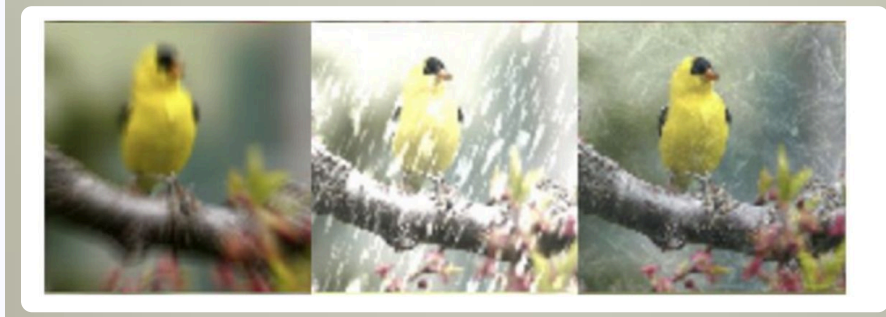
- The digit “8” is shown in different colours (yellow vs orange).
- Colour is **irrelevant** to digit recognition - only the shape matters.
- A robust classifier should **still predict “8”** in both cases.
- This ensures the model is not fooled by meaningless variations.

Good models should focus on the essential feature (digit shape) and ignore irrelevant changes (colour).

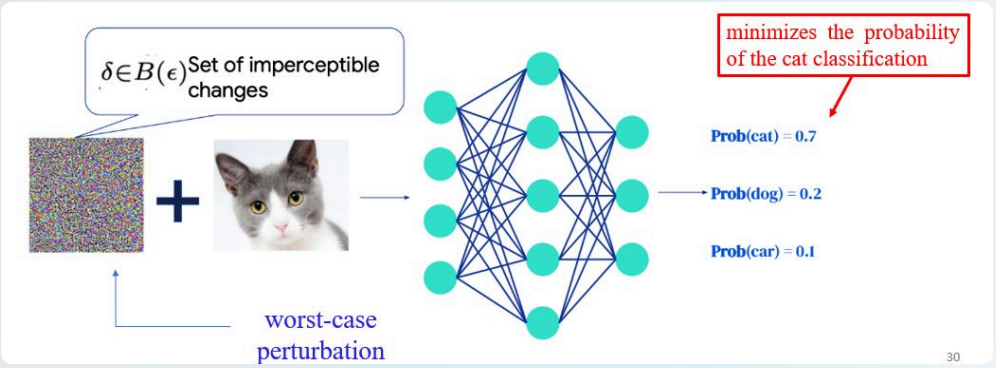


Handling Out-of-Distribution

- A bird classifier is trained on **normal, clean bird images**.
- But in the real world, images may be **blurred, rainy, or noisy**.
- A good model should:
  a. Still recognise the bird when the change is small (robustness).
  b. **Show higher uncertainty** when the image looks too unusual or far from training data (out-of-distribution) → ***If a student studied only math but gets a chemistry question, the right response is "I don't know" instead of guessing with confidence.***



A reliable system is not only robust to common changes (blur, weather, lighting), but also honest enough to say *"I'm not sure"* when it sees something unfamiliar.



How handle Adversarial attack (Defence Strategies)

1) Adversarial Training

- Suppose we have a **cat image**.
- The model correctly predicts:
  - Cat (70%), Dog (20%), Car (10%).
- Now, we add a **tiny, invisible noise pattern** (worst-case perturbation).
  - To humans, the image still **looks like a cat**.
  - But the noise is crafted to **confuse the model**.
- After adding the noise, the model may reduce confidence:
  - Cat (40%), Dog (50%), Car (10%).
  - The model is now tricked into saying it is a **dog**, even though the picture still shows a cat.

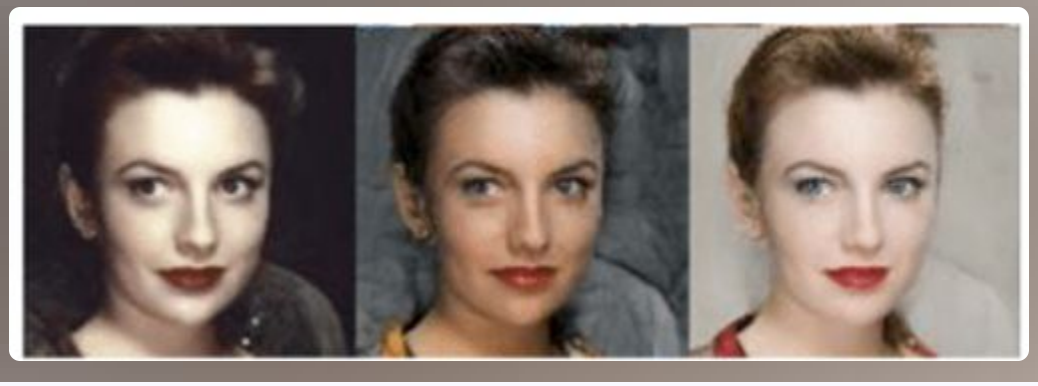
Adversarial training = teach the model during training to also "see through" these noisy images and still predict CAT.

For Example:

Normal question: "2 + 2 = ?" → Student answers **4**.

Trick version: "2 + two = ? (with distractions in wording)" → Student still answers **4**.

Over time, the student learns not to be distracted by tricks.



Reducing Bias with Adversarial Mixing

Neural networks can pick up **unintended correlations** in data (e.g., linking skin tone with smiling) → **algorithmic bias**, where some groups are classified less accurately than others.

Case Study: CelebA Dataset

- Two groups formed based on **skin tone**.
- Task: Predict whether a person is **smiling**.
- Observation: Standard classifier had a **large gap (33.5%)** in accuracy between groups.

Adversarial Mixing

- During training, the model is exposed to **semantic perturbations** (like different skin tones, lighting, or background changes).
- The model is then forced to **focus only on the relevant feature** (smiling vs. not smiling), while ignoring irrelevant features (skin tone).

Result

- Accuracy gap reduced from **33.5% → 3.8%**.
- Demonstrates that adversarial mixing helps create **fairer and more inclusive AI systems**.
- The model should judge **smile vs. no smile**, not skin tone or other irrelevant cues.

Gowal, S., Qin, C., Huang, P. S., Cemgil, T., Dvijotham, K., Mann, T., & Kohli, P. (2020). Achieving robustness in the wild via adversarial mixing with disentangled representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 1211-1220).

Defensive Distillation: Teacher–Student Analogy

Teacher model is trained first, like a teacher who already knows the subject.

- Instead of rigid answers (e.g., *100% Cat, 0% Dog*), the teacher gives nuanced probabilities:
  - This image is very likely a Cat (90%), but it has Dog-like features (8%) and maybe a hint of Fox (2%).

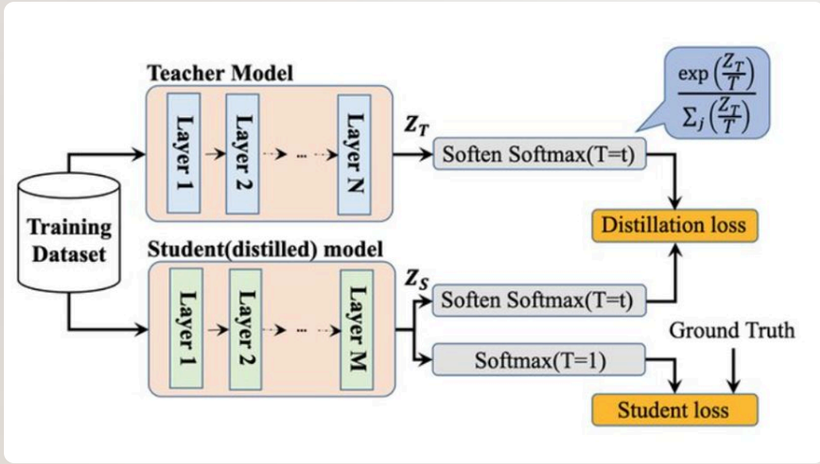
Student model learns from these **soft labels** instead of sharp one-hot labels.

Student model learns **relationships between classes** (Cat vs. Dog vs. Fox), not just hard boundaries.

Prevents **overconfidence** → less chance of confidently wrong predictions.

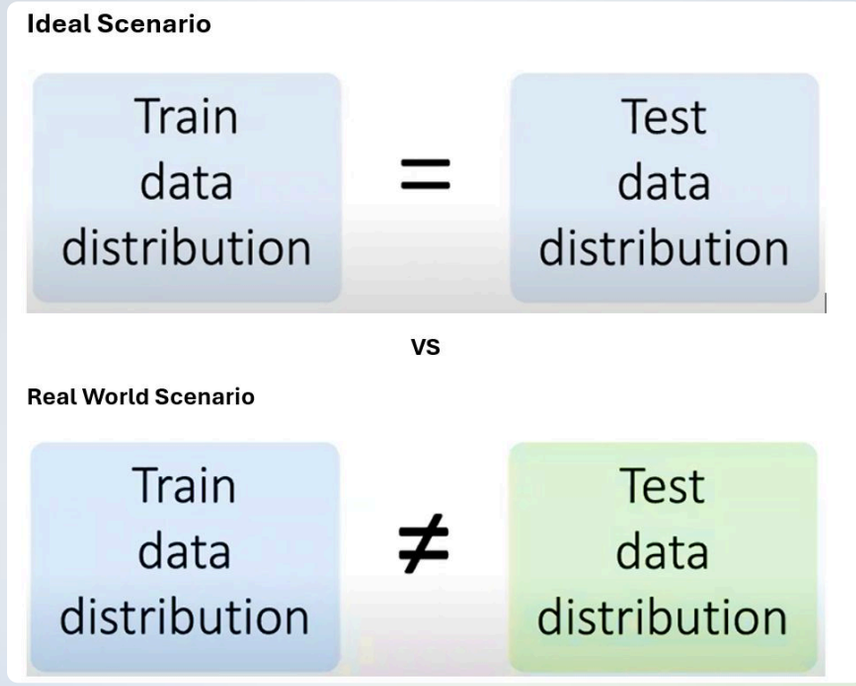
Makes the model **more robust to noise and adversarial attacks**.

Analogy: The teacher not only gives the answer but also explains how sure they are and what other options were possible. The student becomes wiser and less easily fooled.



Standard Assumption in Machine Learning

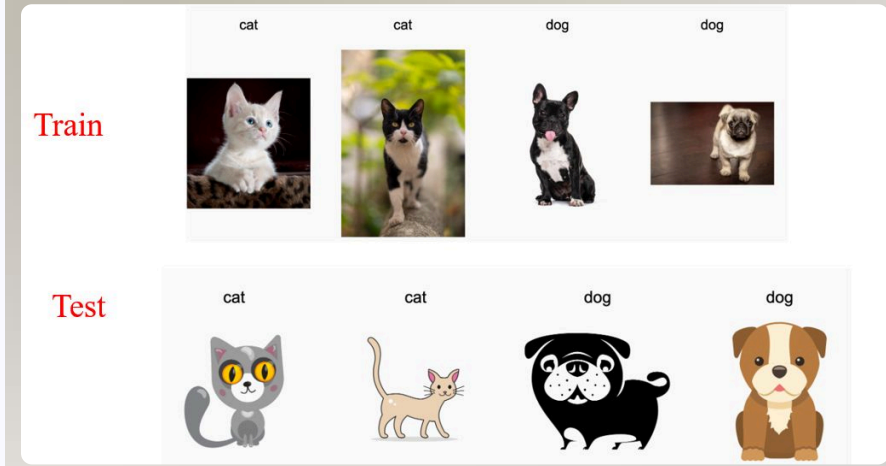
- **Assumption**: Machine Learning models are usually trained with the assumption that training data and test data come from the same distribution.
  - **Example:** If you train a model to recognize cats and dogs using high-quality studio photos, you assume it will also be tested on similar high-quality photos.
- **This assumption works (in ideal cases)**:
  - It allows models to achieve very high accuracy in controlled environment.
- **The problem**:
  - In the real world, this assumption is often broken. Data in deployment may be noisy, adversarially perturbed, or from a shifted distribution (e.g., photos of cats in the wild vs. cats in a studio).
  - This gap between the ideal assumption and real-world conditions is exactly why robustness strategies (like adversarial training, defensive distillation, etc.) are needed.



Distribution Shifts

A distribution shift happens when the data seen during training is **different** from the data seen during testing or deployment.

- **Example**:
  - **Training Data**: Realistic photos of cats and dogs.
  - **Testing Data**: Cartoon drawings of cats and dogs.
  - The labels are the same (“cat” vs “dog”), but the appearance has changed (real vs cartoon).
  - This is an example of **covariate shift** - the input distribution (features) changes, but the task remains the same.
- Distribution shift is one of the *biggest reasons* models fail in the real-world.
- It motivates techniques like **domain adaptation**, **data augmentation**, and **robust training methods** to bridge the gap.



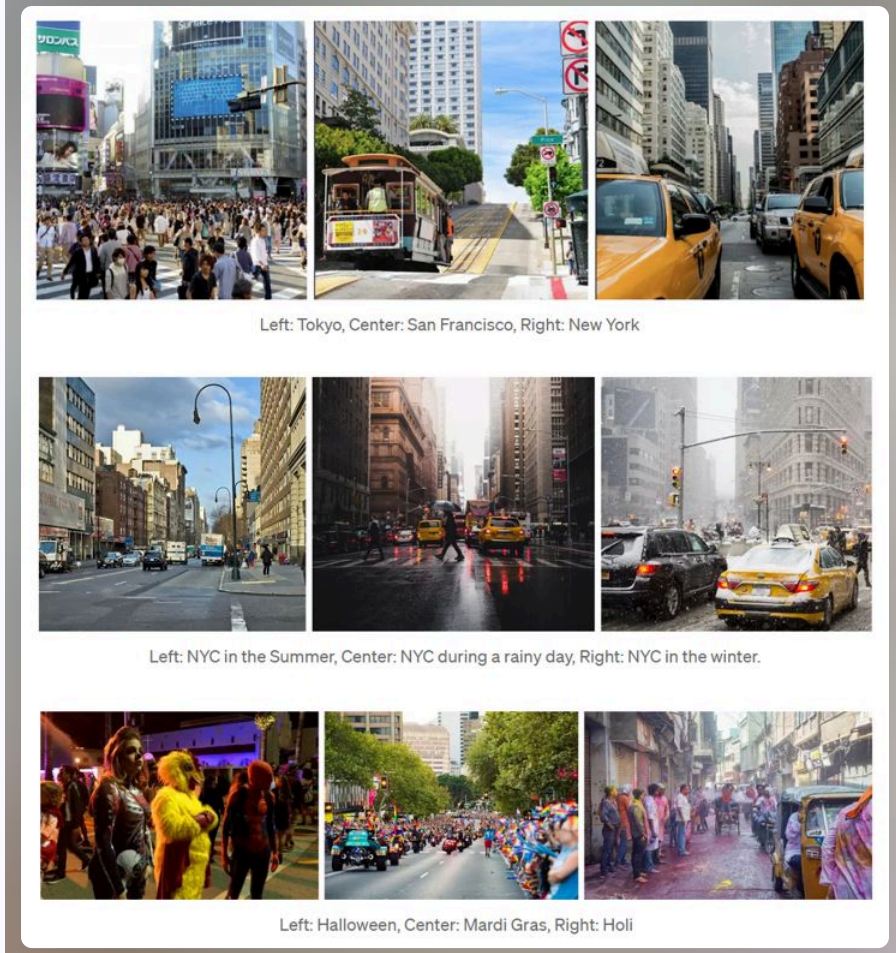
Distribution Shifts in Real-World Example

- **Domain shift** happens when the environment changes between training and testing.

Example differences:

- **Vehicles** (buses in San Francisco, yellow taxis in New York, bicycles in Tokyo)
- **Street signs & markings** (different languages, shapes, and rules)
- **Buildings & houses** (styles vary across cities)
- **People's appearance** (dress code, density, cultural habits)

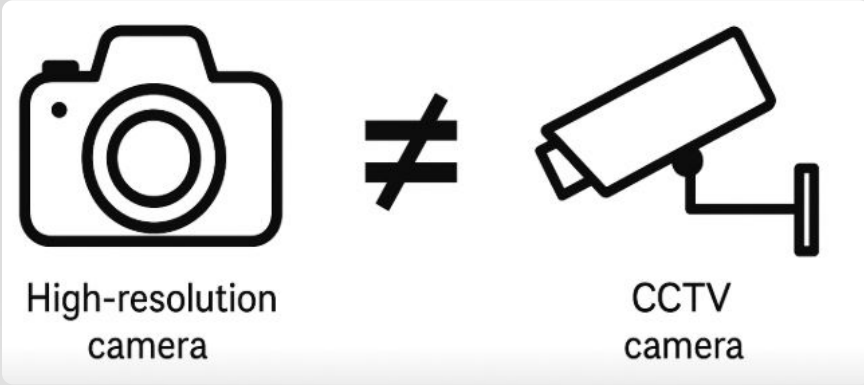
Models trained in one city may **struggle in another** because the input distribution looks different.



Distribution shifts in real-world example

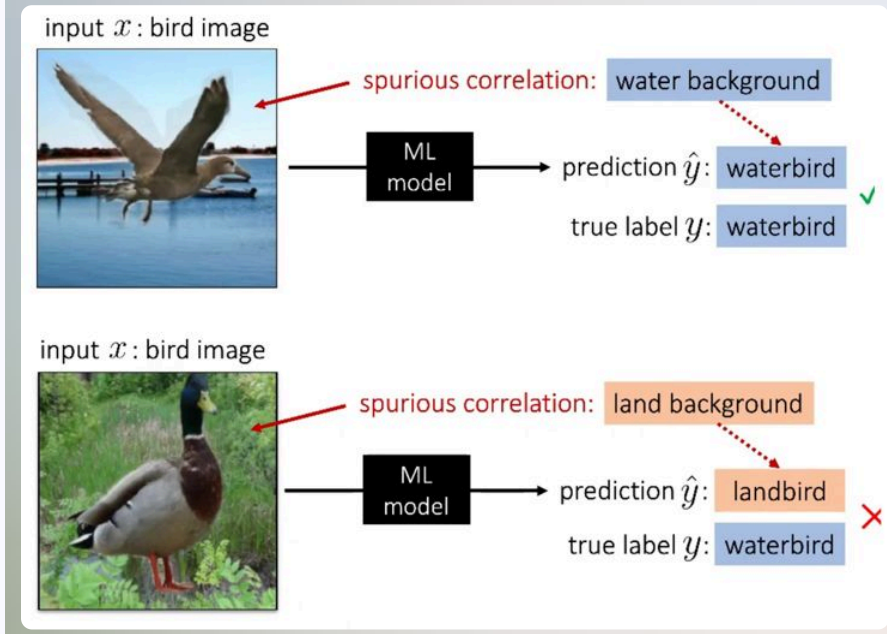
Hardware changes → domain shifts

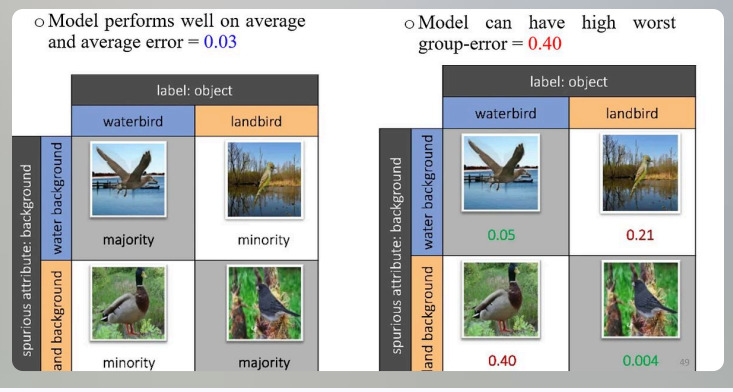
- If you switch to a different camera or sensor, the images it produces might change slightly.
- To us humans, the two images look almost the same.
- But for a **neural network**, even small changes can affect predictions.



Models can latch onto Spurious correlations

- Machine learning models may make predictions based on background cues instead of the actual object.
- **Example:** A model trained to classify ***waterbirds* vs. *landbirds*** might rely on the presence of water in the background rather than features of the bird itself.
- This creates a misleading correlation – the model thinks "water = waterbird."
- It works fine when training and test data have similar backgrounds, but fails when the context changes (e.g., a waterbird in a dry field).
- The model is correct for the **wrong reason**, which makes it unreliable in the real world.





Model performance and fairness across groups

1. **Two labels:**
   - Waterbird
   - Landbird
2. **Two spurious attributes (things the model *should not* rely on but does):**
   - Water background
   - Land background
3. **This creates 4 groups**:
   - **Waterbird in water background** → *majority* (common in training).
   - **Landbird in land background** → *majority* (common in training).
   - **Waterbird in land background** → *minority* (rare in training).
   - **Landbird in water background** → *minority* (rare in training).

- Waterbird + water background → error = **0.05** (very low, good).
- Landbird + land background → error = **0.004** (excellent, almost no mistakes).
- Landbird + water background → error = **0.21** (bad).
- Waterbird + land background → error = **0.40** (terrible).

During training, the model saw many waterbirds in water, landbirds on land.

So it "cheated" and learned background ≈ label.

When background and label disagree, the model gets confused → high error.

How make models more robust to distribution shift?

1. **Monitoring (What happened?)**

Tracks system health and metrics.

Example:

Accuracy drops from 90% to 70% overnight → monitoring flags it.

2. Observability (Why did it happen?)

Lets us debug the root cause by peeking inside the system.

Example:

Discover accuracy drop is because more “landbirds with water backgrounds” appear in test data.

3. Instrumentation (How we achieve it)

Adding sensors/logs to make monitoring & observability possible.

Example:

Add timers around model inference → to know which step is slow.

Log anomalies: e.g., if input image resolution is lower than training data (CCTV vs. DSLR).

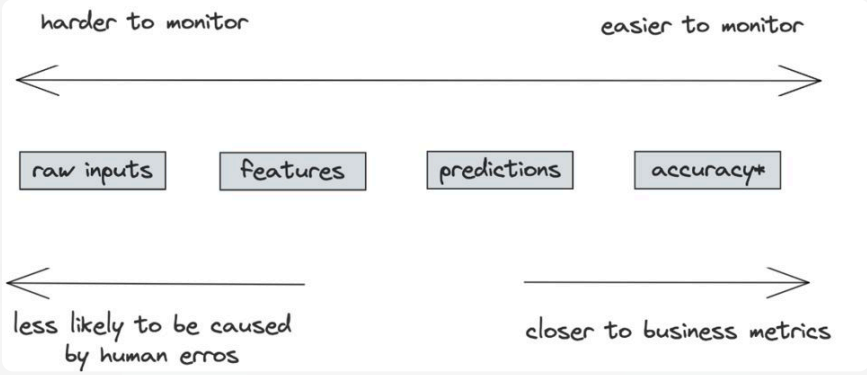
Analogy

- ***Monitoring = fuel gauge in your car (tells you fuel is low).***
- ***Observability = opening the bonnet to see if it’s a leak or just high consumption.***
- ***Instrumentation = actually installing the fuel gauge and sensors.***

What to monitor (the 4 artifacts)

- **Inputs** – Has the data changed?
  - Example: road images are suddenly darker because they are taken at night.
- **Features** – Are the important patterns the model uses changing?
  - Example: colours or edges look different because of poor lighting.
- **Predictions** – Is the model behaving differently?
  - Example: it starts predicting “not a stop sign” more often.
- **Performance/accuracy** – Is the model becoming less correct?
  - Example: accuracy falls from 95% to 75%.

Data changes → features change → predictions change → accuracy may drop



How to Make Models More Robust: Data Augmentations

- **Mixup**:
  Combines two images into one blended image.
  Example: A cat + a dog image = “60% cat, 40% dog.”
  → Helps the model generalise instead of memorising exact details.
- **AutoAugment**:
  Automatically searches for the best set of augmentation strategies.
  Example: Learns that slightly rotating + changing brightness improves accuracy.
- **AugMix**:
  Randomly mixes multiple augmentations together.
  Example: Rotate + colour change + contrast shift all applied at once.
  → Increases robustness against unusual distortions.
- **PixMix**:
  Mixes real images with unrelated images (like patterns).
  Example: A bird image combined with a fractal background.
  → Makes the model less sensitive to noise or background changes.



Summary

- Use of AI in safety-critical applications
- Robustness definition
- Adversarial Example
- How Enabling Safe and Robust AI
- Specification Driven ML
- Adversarial Defence Strategy
- Distribution shifts with real-world examples
- How make models more robust to distribution shift?

Questions

